

Choosing a Method and Reading the Roots

Answer Key

Algebra 1

Quick Answers

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| 1. $-9, 9$ | 7. $-\frac{1}{3}, 2$ |
| 2. $2, 5$ | 8. (a) both solutions = $-9, 5, \text{---}$ |
| 3. $-1, 7$ | 9. $-6, 4$ |
| 4. 1.16 | 10. -0.64 |
| 5. $0, 4$ | 11. $-3, 0, 3$ |
| 6. (a) both solutions = $0, 2, \text{---}$ | 12. (a) both solutions = $6, 18, \text{---}$ |
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What is verified

9 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 6, 8, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSA-CED.A.1 – *problems 8, 12*

HSA-REI.B.4 – *problems 1, 2, 3, 4, 5, 6, 7, 9, 10, 11*

Difficulty 1–5 of 5 across 15 tagged checks.

Common wrong answers

5. *If they got 4: divided both sides by $2x$, discarding the root $x = 0$*

10. *If they got -1.24 : used b instead of $-b$, starting the numerator with -3*

11. *If they got $-3, 3$: divided by x and then solved only x squared minus nine*

1. Solve: $x^2 = 81$

Solution: The squared term is already alone, so square roots are the fastest route — factoring would work but costs an extra step. Root both sides and write the $\pm$ at once.

$$x = \pm\sqrt{81}$$

$x = -9$ or $x = 9$

2. Solve: $x^2 - 7x + 10 = 0$

Solution: Leading coefficient 1 and a quick factor pair, so factoring beats the formula here. Product 10, sum -7 : the pair is -2 and -5 .

$$x^2 - 7x + 10 = 0$$

$$(x - 2)(x - 5) = 0$$

$$\boxed{x = 2 \quad \text{or} \quad x = 5}$$

3. Solve: $(x - 3)^2 = 16$

Solution: The squared part is alone again — do not expand it first. Root both sides, then undo the subtraction.

$$x - 3 = \pm 4$$

$$x = 3 + 4 = 7 \quad \text{or} \quad x = 3 - 4 = -1$$

$$\boxed{x = -1 \quad \text{or} \quad x = 7}$$

4. Use the quadratic formula on $x^2 + 4x - 6 = 0$. Give the larger solution, rounded to two decimal places.

Solution: No factor pair of -6 sums to 4, so the formula is the method. $a = 1$, $b = 4$, $c = -6$, so $-b = -4$ and the discriminant is $16 + 24 = 40$.

$$x = \frac{-4 \pm \sqrt{40}}{2}$$

The larger root takes the plus sign: $(-4 + 6.3246)/2 = 1.1623\dots$, which rounds to $\boxed{1.16}$

5. Solve: $2x^2 - 8x = 0$

Solution: There is no constant term, so a common factor comes out and the zero-product property finishes it. Dividing by $2x$ would delete the root $x = 0$.

$$2x^2 - 8x = 0$$

$$2x(x - 4) = 0$$

$$\boxed{x = 0 \quad \text{or} \quad x = 4}$$

6. A ball is kicked straight up from the ground; its height in feet after t seconds is $h = -16t^2 + 32t$. (a) Solve $-16t^2 + 32t = 0$. (b) Say which solution is the landing time and what the other describes.

Solution (a): Factor out the common factor $-16t$ (no constant term, so this is the quick route).

$$-16t^2 + 32t = 0$$

$$-16t(t - 2) = 0$$

$$\boxed{t = 0 \quad \text{or} \quad t = 2}$$

(b) — instructor-judged. The ball lands at $t = 2$ seconds. Full credit requires explaining that $t = 0$ is not an error: it is the instant of the kick, when the ball is also

at height zero. It solves the equation correctly but does not answer the question that was asked. An answer that discards $t = 0$ as “not a real solution” has the right final time for the wrong reason.

7. Solve: $3x^2 - 5x - 2 = 0$

Solution: The leading coefficient is not 1, but the numbers are small, so grouping is still quicker than the formula. Product $3 \cdot (-2) = -6$, sum -5 : the pair is -6 and 1 .

$$\begin{aligned} 3x^2 - 6x + x - 2 &= 0 \\ 3x(x - 2) + 1(x - 2) &= 0 \\ (3x + 1)(x - 2) &= 0 \end{aligned}$$

$$\boxed{x = -\frac{1}{3} \quad \text{or} \quad x = 2}$$

8. A rectangular garden is 4 metres longer than it is wide and has area 45 square metres, so $w(w + 4) = 45$. (a) Solve for w . (b) State the width and explain why the other solution is rejected.

Solution (a): Expand, move everything to one side, then factor: product -45 , sum $+4$ gives $+9$ and -5 .

$$\begin{aligned} w^2 + 4w &= 45 \\ w^2 + 4w - 45 &= 0 \\ (w + 9)(w - 5) &= 0 \end{aligned}$$

$$\boxed{w = -9 \quad \text{or} \quad w = 5}$$

(b) — instructor-judged. The width is 5 metres. Full credit requires the reason: -9 satisfies the equation perfectly well, but a width is a length, and a length cannot be negative, so the model rejects it. Calling -9 an arithmetic mistake is not the right reason — checking it in the equation shows it works.

9. Solve: $2(x + 1)^2 = 50$

Solution: Divide by 2 first so the squared part stands alone, then take square roots. Expanding would turn a two-step problem into a five-step one.

$$\begin{aligned} (x + 1)^2 &= 25 \\ x + 1 &= \pm 5 \end{aligned}$$

$$\boxed{x = -6 \quad \text{or} \quad x = 4}$$

10. Use the quadratic formula on $5x^2 - 3x - 4 = 0$. Give the smaller solution, rounded to two decimal places.

Solution: Nothing factors here, so the formula it is. $a = 5$, $b = -3$, $c = -4$, so $-b = +3$ and the discriminant is $9 + 80 = 89$.

$$x = \frac{3 \pm \sqrt{89}}{10}$$

The smaller root takes the minus sign: $(3 - 9.4340)/10 = -0.6434\dots$, which rounds to

-0.64

11. Solve: $x^3 - 9x = 0$

Solution: A cubic, but every term contains x , so factor rather than reach for the quadratic formula. The leftover is a difference of squares, so there are three roots, not two.

$$\begin{aligned}x^3 - 9x &= 0 \\x(x^2 - 9) &= 0 \\x(x - 3)(x + 3) &= 0\end{aligned}$$

$$\boxed{x = -3, \quad x = 0, \quad x = 3}$$

12. Synthesis challenge. With ticket price x dollars, a drama club's profit is $P = -x^2 + 24x - 108$ dollars. (a) Solve $-x^2 + 24x - 108 = 0$. (b) Explain what the two prices mean and which prices give a profit above zero.

Solution (a): Multiply through by -1 first so the leading coefficient is positive, then factor: product 108, sum -24 gives -6 and -18 .

$$\begin{aligned}-x^2 + 24x - 108 &= 0 \\x^2 - 24x + 108 &= 0 \\(x - 6)(x - 18) &= 0\end{aligned}$$

$$\boxed{x = 6 \quad \text{or} \quad x = 18}$$

(b) — instructor-judged. A full answer says that \$6 and \$18 are the break-even prices, where the profit is exactly zero, and that every price strictly between them earns a positive profit because the parabola opens downward (the coefficient of x^2 is negative). Naming the two prices without describing what happens between them is partial credit. Students who add that the best price is halfway between, at \$12, have gone beyond what was asked and are right.